AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

What type is False and how does it differ from "False"?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Translate stated requirements into comparisons. Try the recall before opening notes.

Your learning target

I can translate stated requirements into **comparison**s.

I can combine **Boolean** values with and or and not.

I can choose a case that distinguishes two conditions.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Translate stated requirements into **comparison**s

Combine **Boolean** values with and or and not

Choose a case that distinguishes two conditions

Retrieve what you already know

> What type is False and how does it differ from "False"?

My starting explanation

This week's end goal

Trace **boolean** conditions and explain the changed program before running it.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Boolean conditions case. Separate observations, given facts and assumptions before choosing an explanation.

A fictional activity must take at most **12 minutes** and be quiet. An activity taking exactly **12 minutes** meets the time requirement. A quiet 15-minute activity does not.

Before writing code, underline each requirement and decide whether all requirements must hold or whether alternatives are allowed.

Use a fresh conditions.py file with nonnegative integer minutes and an actual **Boolean** quiet flag. A **comparison** produces True or False for these values. The expression minutes <= 12 includes 12; minutes < 12 excludes it.

Use == for equality and != for inequality. A single = assigns a value to a name; it does not ask whether two values are equal.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Boolean conditions. Connect each labelled input, deciding step and result.

Two requirements must both hold

INPUT CASE	TIME CHECK	COMBINED RESULT
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

```
minutes = 12
quiet = False
time_ok = minutes <= 12
suitable = time_ok and quiet
print(time_ok)
print(suitable)
```

The script prints True then False. The first result concerns time only. The second requires time_ok and quiet. These are separate claims.

Trace each **input** before explaining the combined result; a result without its conditions can be misleading.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Boolean conditions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Change age to 16.

Predicted result

Prediction: The result becomes False.

Why the result follows

Reason: The lower condition remains True, but the upper condition fails.

Words that unlock the idea

> Use the vocabulary of Boolean conditions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Comparison	An expression that checks a relation between values
Boolean	A type with the values True and False
Conjunction	A requirement that both conditions hold
Inclusive or	A condition satisfied when either or both parts hold
Counterexample	A case that shows a claim or rule is wrong

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Boolean conditions, predict a step, then compare with its explanation.

A	B	A and B	A or B
False	False	False	False

A	B	A and B	A or B
False	True	False	True
True	False	False	True
True	True	True	True

Worked explanation · Boolean conditions

With **Boolean** operands, and needs both parts True. **Inclusive or** needs at least one True, so both True is allowed. not reverses a **Boolean**: not True is False and not False is True.

Python can also apply and and or to other types; keep operands **Boolean** here so this table directly describes the result.

A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Boolean conditions

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

The facts we will use

Supplied facts

Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

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Boolean conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check lower boundary

15>=12 is True.



2. Check upper boundary

15<=15 is also True.



3. Combine with and

Both comparisons are True, so allowed is True.

CHANGE ONE THING

Change age to 16.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check lower boundary	15>=12 is True.
2. Check upper boundary	15<=15 is also True.
3. Combine with and	Both comparisons are True, so allowed is True.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

What this example does not establish

Keep the stated limit

This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Keep the stated limit

Read the labelled picture: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Keep the stated limit

Change one thing: Change age to 16. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Boolean conditions claim and explain a repair using the supplied evidence.

```
minutes = 12
quiet = True
print((minutes >= 0) and (minutes <= 12))
print(not quiet)
print(minutes == 12)
```

Check the claim against the facts

This prints True, False and True. Parentheses make the two range checks clear. To express minutes equal to 8 or 12, write (minutes == 8) or (minutes == 12).

Do not write minutes == 8 or 12: the number 12 is not a second equality **comparison**. Later lessons examine truth testing of other types.

The rule uses values supplied by a person. It does not verify that the activity is really quiet, and a True result grants no permission to act.

Our **input** contract already assumes a valid integer and **Boolean**; handling wrong types is a separate task.

Locate the earliest unsupported step

Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

> **State one repair and the checks you would rerun.**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Boolean conditions with support, then explain the deciding step in your own words.

Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

> **A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.**

Explain your prediction to a partner before comparing results.

Practise with less help: Trace separate checks

Use workbook W2 for the connected example “**Boolean** conditions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Boolean conditions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

```
minutes = 12
quiet = False
time_ok = minutes <= 12
suitable = time_ok and quiet
print(time_ok)
print(suitable)
```

```
minutes = 12
quiet = True
print((minutes >= 0) and (minutes <= 12))
print(not quiet)
print(minutes == 12)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Boolean conditions. Explain why each step is justified.

Explain → apply → challenge

Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Testing only a quiet 8-minute activity cannot distinguish and from or because both pass. Use a case where exactly one requirement holds.

Keep expected results beside observed results and explain which requirement caused failure. Next week a program will choose an action using a condition.

> **For the original at-most-12-minutes and quiet rule, choose one input that reveals changing <= to < and a different input that reveals changing and to or. State the expected correct result for each.**

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Boolean conditions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Two requirements must both hold

INPUT CASE	TIME CHECK	COMBINED RESULT
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Combine requirements without replacing AND

> Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

Choose a case that detects replacing and with or between the range check and available.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Boolean conditions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

By the end, I can	How I will show it
Translate stated requirements into comparisons	Explain the deciding step, show a trace or test, and state one limit.
Combine Boolean values with and or and not	Explain the deciding step, show a trace or test, and state one limit.
Choose a case that distinguishes two conditions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Boolean conditions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Translate stated requirements into **comparison**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both **comparison**s are True, so allowed is True.

1. Check lower boundary

2. Check upper boundary

3. Combine with and

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Combine with and Both **comparison**s are True, so allowed is True. A **Boolean** condition represents a yes-or-no result.

And requires both conditions; checking only one boundary would accept values outside the intended interval.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change age to 16. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The result becomes False.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The result becomes False. The lower condition remains True, but the upper condition fails.

Repair a missing building block

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python’s grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Boolean conditions graphic using its labels, deciding evidence and relevant limitation.

01. Boolean conditions: the case

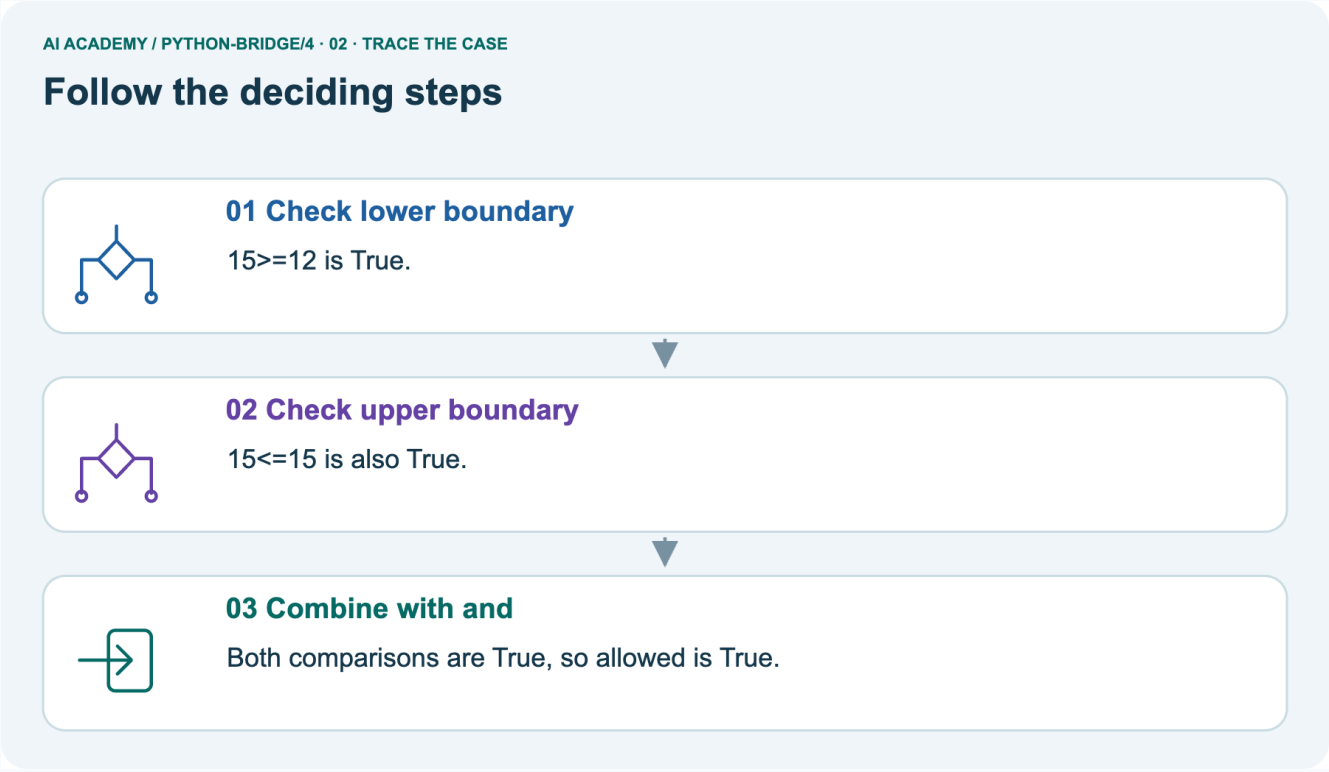


Follow the labelled picture with the matching explanation. Open the vector diagram

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive. Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

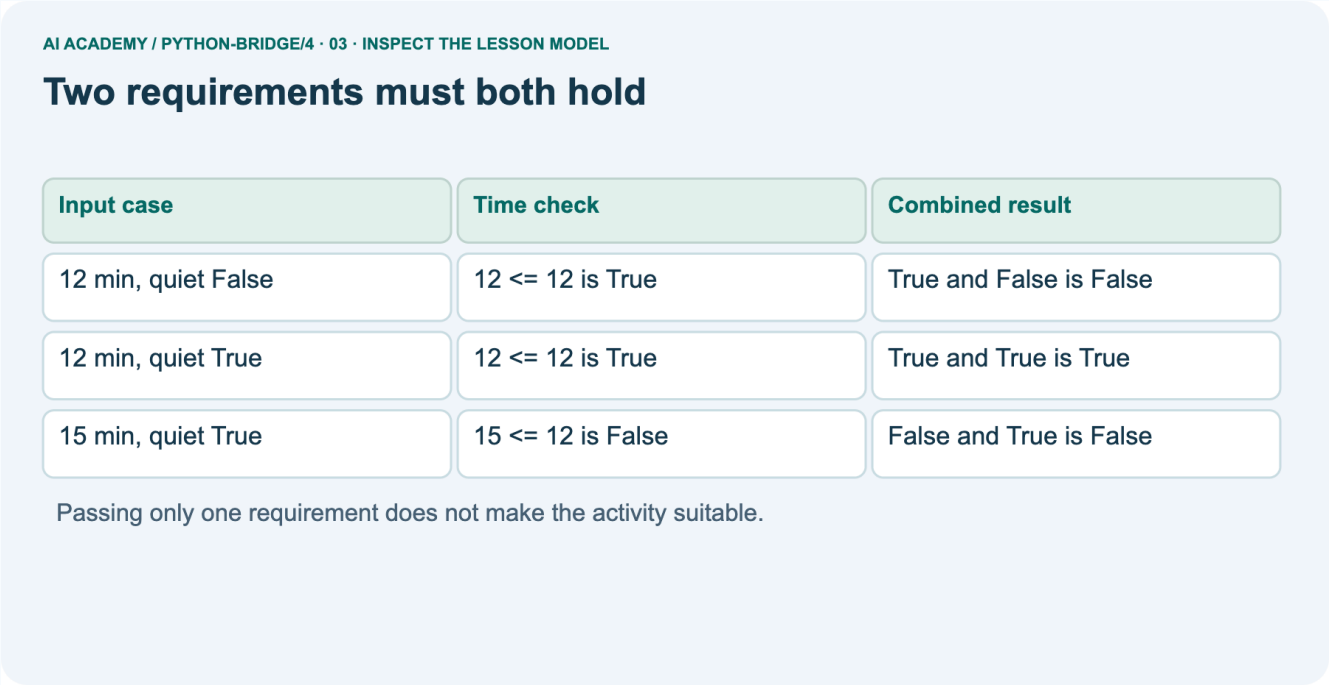


Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two requirements must both hold

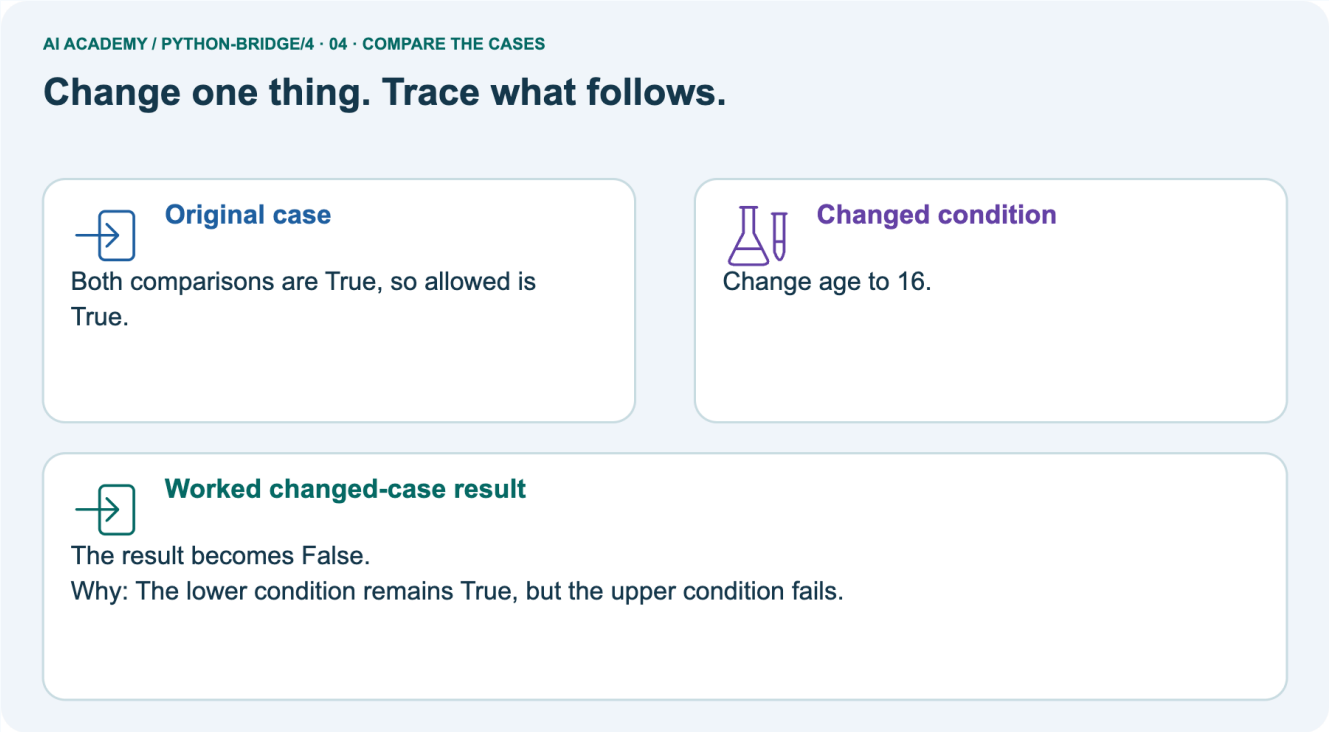


Follow the labelled picture with the matching explanation. Open the vector diagram

Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

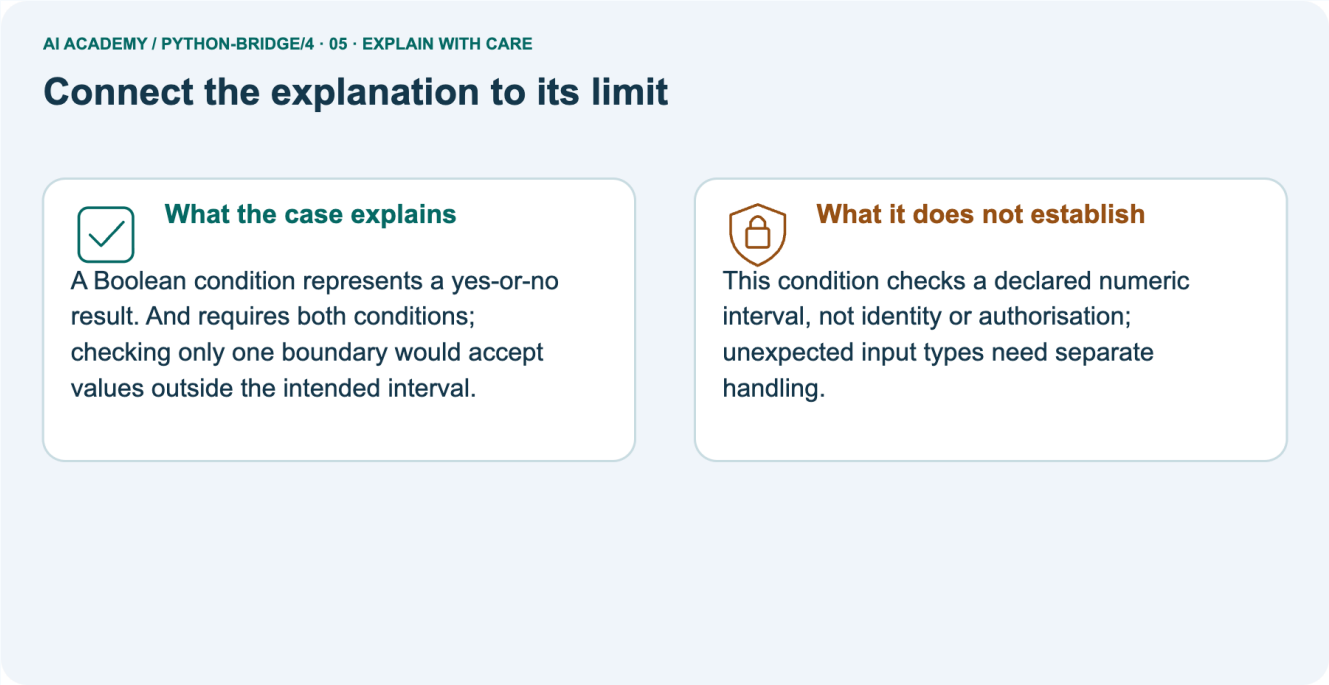


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



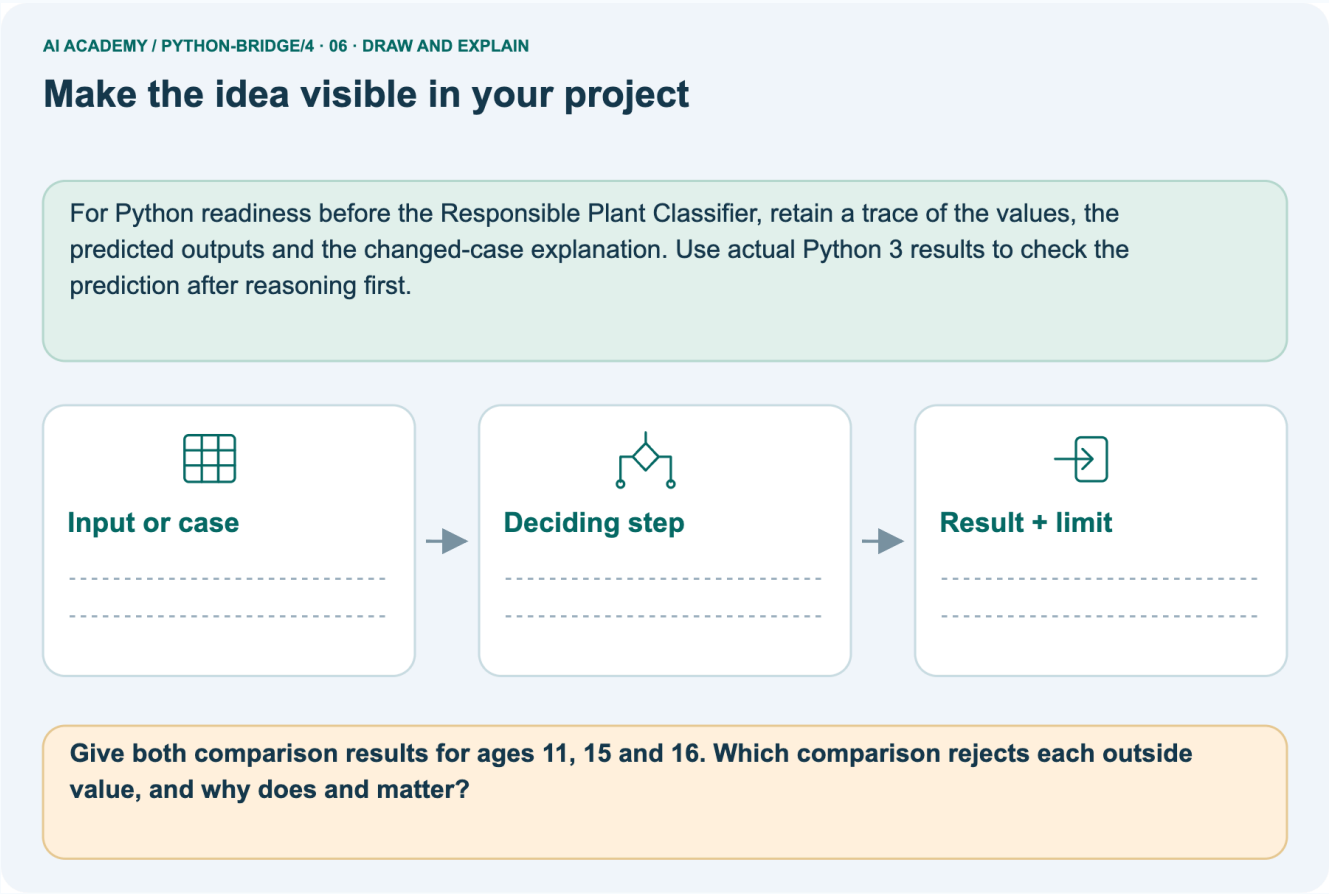
Follow the labelled picture with the matching explanation. Open the vector diagram

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.